

CODE:

```
clc;
clear;
close all;
obj=imread('ball3.jpg');
imshow(obj)
%Segment image
%%divide the image "obj" into its respective RGB intensities
red=obj(:,:,1);
green=obj(:,:,2);
blue=obj(:,:,3);
figure(1)
subplot(2,2,1);imshow(obj);title('original image');
subplot(2,2,2);imshow(red);title('red plane');
subplot(2,2,3);imshow(green);title('green plane');
subplot(2,2,4);imshow(blue);title('blue plane');
%threshold the blue plane
figure(2)
level=0.37;
bw2=imbinarize(blue,level);
subplot(2,2,1);imshow(bw2);title('blue plane thershold');
%remove noise
%fill any holes
fill=imfill(bw2,'holes');
subplot(2,2,2);imshow(fill);title('holes filled');
%remove any blobs on the border of the image
clear = imclearborder(fill);
subplot(2,2,3);imshow(clear);title('remove blobs on border');
%remove blobs that are smaller than 10pixels across
se=strel('disk',10);
open=imopen(fill,se);
subplot(2,2,4);imshow(open);title('remove small blobs');
% Measure diameters of all detected objects
stats = regionprops(open, {'EquivDiameter','MajorAxisLength'});
diameters_equiv = [stats.EquivDiameter];
%show result
figure(3)
imshow(obj)
d=imdistline;%include a line to physically measure the ball
% Extract diameter (each ball = one value)
disp('Diameter of ball (in pixels):');
disp(diameters_equiv);
```